Data Flow Diagram

Treavue — Personalized Travel Itinerary Planner

Project documentation prepared for portfolio and repository use

System context

Traveler → React/Vite frontend → FastAPI backend → Supabase trips table. The itinerary API uses the backend's available destination and activity data. The chat feature is a separate existing capability.

Flow 1 — Generate itinerary

- User enters preferences in frontend.
- Frontend sends request to FastAPI itinerary endpoint.
- Backend validates inputs and builds itinerary from available schedules/data.
- Backend returns structured itinerary.
- Frontend displays the result.

Flow 2 — Save trip

- Frontend submits trip data to FastAPI trip API.
- Backend maps ID, name/title, destination, and full trip payload.
- Supabase stores the full trip object in JSONB.
- Backend returns success or a useful error.
- Frontend retains local browser data as a fallback when configured to do so.

Flow 3 — Read/edit/delete

Frontend requests trip list or a specific trip; updates are sent to the API; delete requests remove the selected record from the configured table.

Security note

The current demo storage integration must not be exposed publicly with a privileged backend key until authentication, verified identity, authorization, and suitable RLS policies are fully enforced end-to-end.